Tenzorske metode za Recommender Systems

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Social Tagging Recommender Systems

- Problem: korisniku preporučiti oznake za određeni artikl

$$Y : U \times I \times T \to \mathbb{R}$$

- $S \subseteq U \times I \times T$

$$D_S := \{(u, i, t_A, t_B) : (u, i, t_A) \in S \wedge (u, i, t_B) \notin S\}$$

Bayesian Personalized Ranking

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$$\Rightarrow \ln(p(\Theta| >_{u,i})) = \sum_{(u,i,t_A,t_B)\in D_S} \ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta)) - \lambda_\Theta ||\Theta||_F^2$$

Stohastički gradijentski spust

$$\ln(p(\Theta| >_{u,i})) = \sum_{(u,i,t_A,t_B)\in D_S} \ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta)) - \lambda_\Theta ||\Theta||_F^2$$

- Problem: D_S brzo raste
- Uniformno biramo $(u, i, t_A, t_B) \in D_S$

$$\nabla ln(p(\Theta| >_{u,i})) \approx \nabla \Big[\ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta) - \lambda \|\Theta\|_f^2 \Big]$$

Model

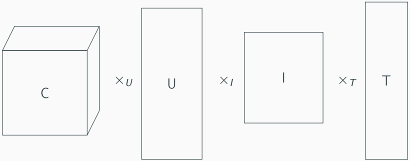
$$\nabla ln(p(\Theta| >_{u,i})) \approx \nabla \Big[\ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta) - \lambda \|\Theta\|_f^2 \Big]$$

$$Y : U \times I \times T \to \mathbb{R} \quad \rightsquigarrow \quad \hat{y} \in \mathbb{R}^{|U| \times |I| \times |T|}$$

$$y_{u,i,t_A,t_B} := \hat{y}_{u,i,t_A} - \hat{y}_{u,i,t_B}$$

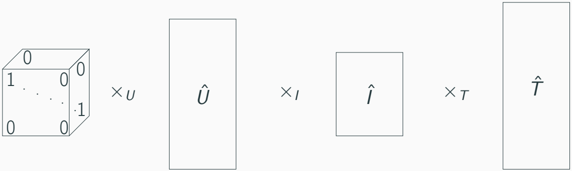
$$\Theta = (C, U, I, T)$$

Tuckerova faktorizacija



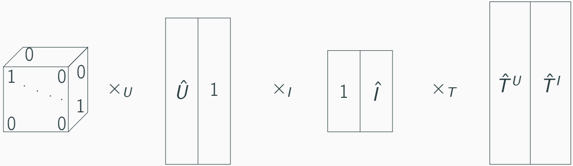
$$\hat{y}^{TD}_{u,i,t} = \sum_{u'} \sum_{i'} \sum_{t'} \hat{c}_{u',i',t'} \hat{u}_{u,u'} \hat{i}_{i,i'} \hat{t}_{t,t'}$$

$$\hat{C} \in \mathbb{R}^{k_u \times k_i \times k_t}, \qquad \hat{U} \in \mathbb{R}^{|U| \times k_u}$$
$$\hat{I} \in \mathbb{R}^{|I| \times k_i}, \qquad \hat{T} \in \mathbb{R}^{|T| \times k_t}$$



$$\hat{y}^{CP}_{u,i,t} = \sum_{f}^{k} \hat{u}_{u,f}\,\hat{i}_{i,f}\,\hat{t}_{t,f},$$

PITF



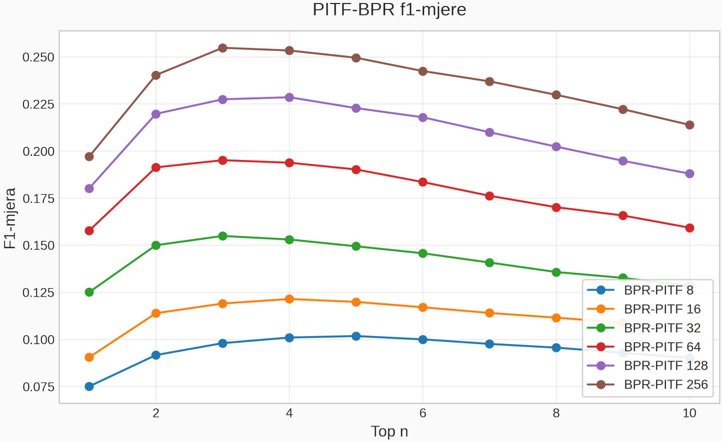
$$\hat{y}_{u,i,t} = \sum_f \hat{u}_{u,f}\hat{t}^U_{t,f} + \sum_f \hat{i}_{i,f}\hat{t}^I_{t,f} + \sum_f \hat{u}_{u,f}\hat{i}_{i,f}$$

$$\hat{U} \in \mathbb{R}^{|U|\times k}, \qquad \hat{I} \in \mathbb{R}^{|I|\times k}$$
$$\hat{T}^U \in \mathbb{R}^{|T|\times k}, \qquad \hat{T}^I \in \mathbb{R}^{|T|\times k}$$

MovieLens

- 7-core dataset
- 4,304 korisnika, 7,475 filmova, 19,745 oznaka, 4,012,200 veza
- korisnički generirane oznake

Utjecaj parametra k - MovieLens



Utjecaj parametra λ - MovieLens

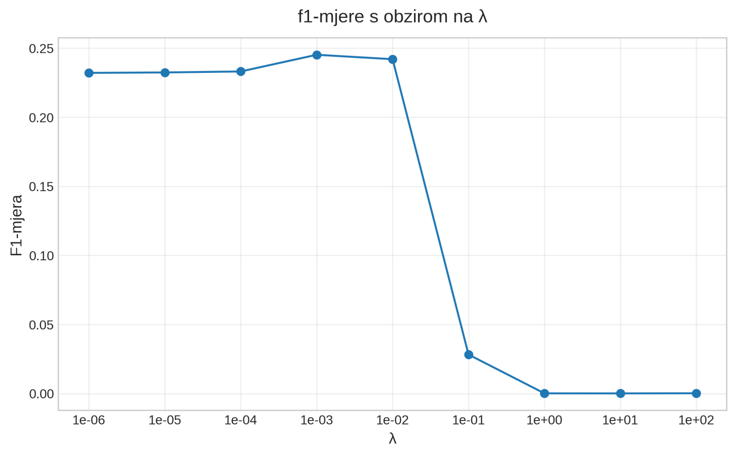
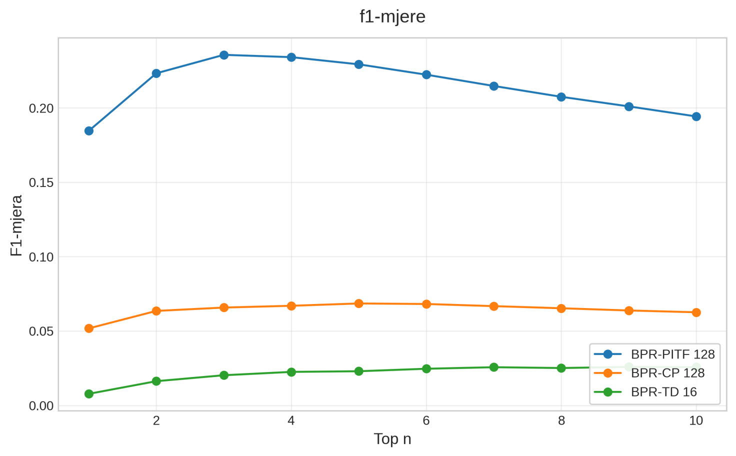


Figure 1: Caption

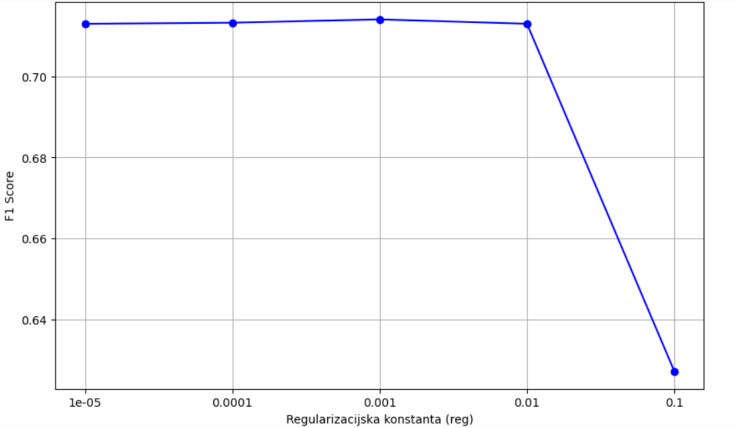
Usporedba modela - MovieLens



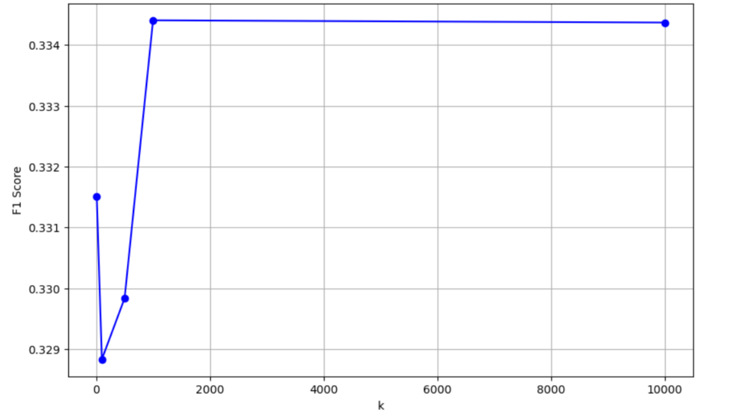
libimseti

- 10-core dataset
- 17,000,000 ocjena
- 3 oznake (1-3, 4-6, 7-10)

PITF - libimseti



CP - libimseti



Delicious

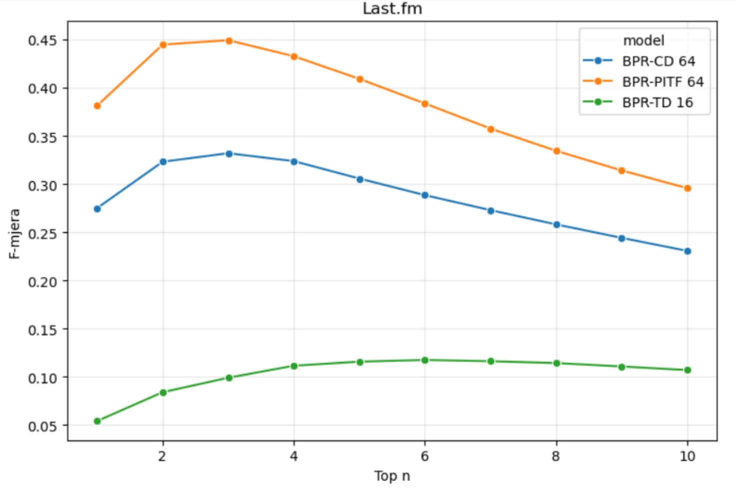
- 3-core dataset
- 1,458 korisnika, 5,074 web-dokumenata, 4,233 oznaka, 84,000 veza

Algoritam	F1@3
PITF	0.22
CP	0.04

last.fm

- 7-core dataset
- 811 osoba, 2,386 glazbenika/grupa, 1,288 oznaka 111,149 veza

Usporedba modela - last.fm



Utjecaj parametra k - last.fm